Jinzhao Lian

Number: +86 13155805132 | Email: 2023201891@ruc.edu.cn

RESEARCH INTERESTS

My research interests lie in efficient and scalable systems for foundation models, with a focus on algorithm-system co-design for LLM serving on heterogeneous GPU clusters.

EDUCATION

Renmin University of China **Beijing, China**
BSc Statistics and Data Science *Expected June 2027*

- **Core Courses:** Deep Learning (96), Parallel Computing (91), Reinforcement Learning (93), Machine Learning (90), Mathematical Statistics (95), Optimization Methods. GPA：3.63/4.0
- **Scholarship &Honors:** China Scholarship Council (CSC) State-Sponsored Scholarship (2025). Merit Student Award (2024&2025), Academic Progress Scholarship (2024). CUMCM 2024 (National 2nd Prize), MCM/ICM 2025 (Meritorious Winner).

PUBLICATIONS

- X. He*, Z. Fang*, **J. Lian***, D. H.K. Tsang, B. Zhang, Y. Chen. FREESH: Fair, Resource- and Energy-Efficient Scheduling for LLM Serving on Heterogeneous GPUs. *Under submission to IEEE TPDS.* (**These authors contributed equally.*) https://arxiv.org/abs/2511.00807
- Z. Dan, J. Zhang, **J. Lian**, Y. Chen, S. Liu. ToolPulse: Side-Channel Eavesdropping on Tool-Using LLM Agents. *Under submission to USENIX Security.*

RESEARCH EXPERIENCE

University of Alberta (Efficient LLM Serving Systems) **Edmonton, Canada**
Summer Research Intern / Supervisor: Yize Chen *June 2025 – October 2025*

- Adaptive Scheduling: Designed the Least-Laxity-First (LLF) scheduling logic. Quantified dynamic request urgency (laxity) to resolve the HOL blocking issue in heterogeneous clusters, cutting SLO violations by 62.8% versus FCFS under high-concurrency.
- Predictive Modeling: Fine-tuned a BERT classifier to predict LLM output token lengths and trained an LSTM network on Azure traces to forecast traffic peaks for proactive resource scaling.
- Resource Allocation: Reduced over-provisioning by re-optimizing cluster-wide GPU allocation and TP configurations every 30 minutes via MILP.
- System Implementation: Integrated the scheduler and predictor into a router interacting with vLLM inference engines. Achieved 28.6% energy savings on production workloads.

The University of Hong Kong (Efficient Agent) **Hong Kong, China**
Student Research Assistant / Supervisor: Shinan Liu May *2026 – August 2026*

- Developed a latency-free intent-drift detection method, achieving 40.2% exact step localization accuracy on the RootSE development set without additional LLM calls.
- Supported ToolPulse's cross-model evaluation by running Terminal-Bench 1 and 2 workloads across 11 LLM backends in Codex and Claude Code.

ADDITIONAL INFORMATION

Languages: Mandarin (native); English (fluent, TOEFL iBT:100/120).
Leadership: Vice Chair, Statistical Investigation Association; Class Representative.
Interests: Badminton, Skiing, Fitness, Documentaries.